Nonreciprocal spin couplings and the Doppler breakdown of driven chiral patterns

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An electric current, or any designed nonreciprocal coupling with a fixed direction, enters the stationary equation of a co-moving chiral magnetic pattern only through two vectors of opposite time-reversal parity, and in Landau–Lifshitz form its dissipative part is exactly $(\beta-\alpha)\mathbf{u}$, independent of the Hall motion of the pattern. The residual magnetic group of this structure fixes the first-order response: a helix acquires a cone magnetization $M = su_{\rm eff}/D$, a pattern-mediated Edelstein effect, and a skyrmion lattice a dipolar distortion perpendicular to the reactive vector rather than to the current. A driven pattern has no precessional Walker breakdown, because the restoring force of its conjugate variable does not saturate; its static branch ends instead in a Landau instability of its own Goldstone mode, $\omega_0(k) < \lambda u_{\rm eff} k$, in which the nonadiabatic parameter enters only through $u_{\rm eff} = (\beta/\alpha - 1)u$. The order of the Goldstone dispersion decides the threshold, which is finite for a helix, $u_L = D/\sqrt{2}s$, and vanishes as $1/L$ for skyrmion lattices and for stripes driven at any angle to their wave vector. The Doppler efficiency λ is a static property of the pattern, one minus the counterflow momentum carried by strain-induced spin canting, and falls from 1 for continuous phase textures to 0.77–0.11 for skyrmion lattices of increasing field; below threshold the same efficiency puts the lattice phonons in a moving Gibbs ensemble, and exchange-even couplings such as damping-like spin–orbit torque pump them through the dual mechanism. We thus show that the fate of a current-driven chiral pattern is set by the order of its Goldstone dispersion and by a single static number, the Doppler efficiency of its internal relaxation.

I. INTRODUCTION

Chiral magnets form spontaneous patterns—helices, stripes and skyrmion lattices—whose period is set by the competition of exchange and the Dzyaloshinskii–Moriya interaction (DMI) [1, 2]. An electric current moves these patterns through the spin-transfer torques of Zhang and Li [3, 4], at current densities far below those needed for domain walls [5, 6], and Fig. 1 shows the three patterns studied here together with the first effect of a current on each. Two questions about this driven motion remain open. First, a current is a vector with a fixed direction, so it lowers the symmetry of the pattern; which distortions does it then allow, and which does it forbid? Second, a domain wall driven too hard undergoes Walker breakdown [7]; what is the corresponding end of a driven pattern, and what sets its threshold?

Partial answers exist for special cases. A longitudinal current destabilizes a skyrmion string by the emission of Goldstone spin waves once $\omega(k_z) < (\beta/\alpha - 1)v_s k_z$ [8], and a current perpendicular to the wave vector of a helix destabilizes it transversely [9]. A uniform current exerts no torque on a sixfold skyrmion lattice [10], and the cone induced in a helix reverses with its chirality [11]. These results use different models and different language. They leave open whether a current can destabilize the in-plane phonons of a skyrmion lattice at all; the string criterion assumes a phase that advances continuously, while a rigid-particle (Thiele) picture of the lattice gives no instability.

Here we treat the current as what it is on a lattice: the antisymmetric part of a nearest-neighbour nonreciprocal spin coupling, $J_{ij} \neq J_{ji}$, with a fixed direction [12]. In the frame co-moving with a pattern, such a fixed bias enters the stationary equation only through two vectors of opposite time-reversal parity (Sec. II). This one structure gives three results. The residual magnetic group fixes the first-order response, with nonsymmorphic protection of the forbidden components (Sec. III). A pattern has no precessional Walker breakdown; its static branch ends in a Landau instability of its own Goldstone mode, whose threshold is finite or zero according to the order of the dispersion (Secs. IV and V). And the strength of the coupling between the current and that mode is a static property of the pattern, the Doppler efficiency λ, set by the counterflow of strain-induced spin canting (Sec. VI). Below threshold the same λ defines a moving Gibbs ensemble of the lattice phonons (Sec. VII), and couplings that a current cannot produce, such as nonreciprocal DMI, enter the same framework through a dual mechanism (Sec. VIII). Section IX states where the model's assumptions hold in real samples and lists the predictions as dependencies. What a pattern does beyond its threshold is the subject of a companion paper [13].

II. THE CO-MOVING STATIONARY EQUATION

We use the Landau–Lifshitz–Gilbert equation with the Zhang–Li torques [3, 4],

$$\partial_t \mathbf{m} = -\mathbf{m}\times\mathbf{H} + \alpha\,\mathbf{m}\times\partial_t\mathbf{m} - (\mathbf{u}\cdot\nabla)\mathbf{m} + \beta\,\mathbf{m}\times(\mathbf{u}\cdot\nabla)\mathbf{m}, \tag{1}$$

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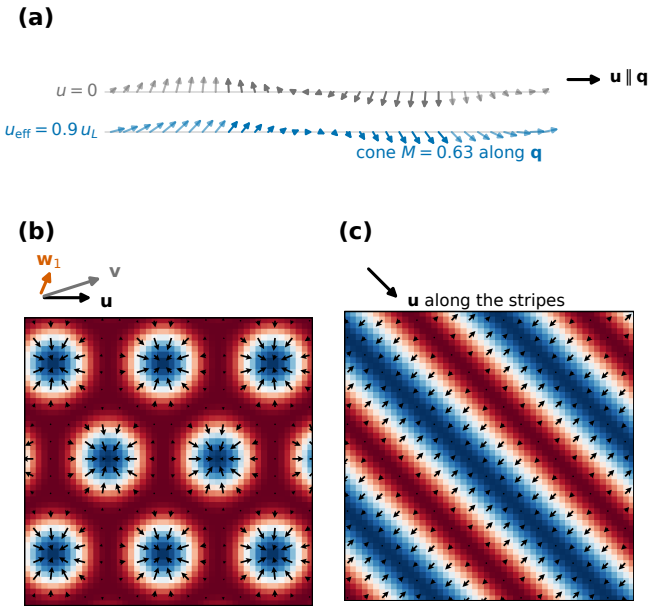


FIG. 1. The three driven chiral patterns and the first-order effect of a current on each. (a) Bloch helix (every second spin of one period, oblique view): unbiased, and at $u_{\rm eff} = 0.9\,u_L$, where the spins tilt into a cone $M = su_{\rm eff}/D$ along $\mathbf{q}$ [Eq. (6)]. (b) Néel skyrmion lattice at $B = 0.5D^2/J$ (colour m_z from -1, blue, to $+1$, red; arrows the in-plane spin) with the current $\mathbf{u}$, the Hall velocity $\mathbf{v}$ and the reactive vector $\mathbf{w}_1 = \mathbf{v} - \mathbf{u}$ from the exact Thiele projection ($\alpha = 0.3$, $\beta = 0.6$). (c) Diagonal Néel stripes (period $P = 30$) with a current along the stripes, the direction that couples to their bending mode.

with $\mathbf{H} = -\delta E/\delta\mathbf{m}$ for exchange J, interfacial (Néel) or bulk (Bloch) DMI D and a field $B\hat{z}$; $\mathbf{u}$ is the spin-drift velocity, α the Gilbert damping and β the nonadiabatic parameter. Units are $J = 1$, $|\mathbf{m}| = 1$, gyromagnetic ratio 1 and lattice constant 1; s denotes the spin density, $s = 1$ in the numerics. For a pattern moving rigidly with velocity $\mathbf{v}$, $\partial_t\mathbf{m} = -(\mathbf{v}\cdot\nabla)\mathbf{m}$, and Eq. (1) becomes the stationary condition

$$0 = -\mathbf{m}\times\mathbf{H} + (\mathbf{w}_1\cdot\nabla)\mathbf{m} + \mathbf{m}\times(\mathbf{w}_2\cdot\nabla)\mathbf{m},$$
$$\mathbf{w}_1 = \mathbf{v} - \mathbf{u}, \qquad \mathbf{w}_2 = \beta\mathbf{u} - \alpha\mathbf{v}. \tag{2}$$

Four properties follow.

Parity. The exchange and DMI fields are odd in $\mathbf{m}$, so $\mathbf{m} \to -\mathbf{m}$ maps $(\mathbf{w}_1, \mathbf{w}_2) \to (-\mathbf{w}_1, \mathbf{w}_2)$ (a field B is reversed with it). $\mathbf{w}_1$ is a time-reversal-odd polar vector, the reactive slot; $\mathbf{w}_2$ is time-reversal-even, the dissipative slot. Time reversal is therefore a usable symmetry element of the stationary problem even with damping (Sec. III).

Thiele projection. Projecting Eq. (2) on the translation modes gives $G\mathbf{w}_1 + \mathcal{D}\mathbf{w}_2 = 0$ with the gyrotropic tensor G and the dissipation tensor $\mathcal{D}$ [14]; this fixes $\mathbf{v}$ exactly. For one-dimensional patterns $G = 0$, so $\mathbf{w}_2 = 0$, $\mathbf{v} = (\beta/\alpha)\mathbf{u}$ and $\mathbf{w}_1 = u_{\rm eff}\hat{u}$ with

$$u_{\rm eff} = \left(\frac{\beta}{\alpha} - 1\right)u. \tag{3}$$

For a skyrmion lattice $\mathbf{w}_1 \perp \mathbf{w}_2$, and $\mathbf{w}_1$ is rotated from $\mathbf{u}$ by the angle $\arctan(G/\alpha\mathcal{D})$ [Fig. 2(b)].

Only $(\beta-\alpha)\mathbf{u}$ drives. Since $\mathbf{w}_2 = (\beta-\alpha)\mathbf{u} - \alpha\mathbf{w}_1$, Eq. (2) depends on $\mathbf{u}$ and β only through $(\beta-\alpha)\mathbf{u}$ at fixed α. Every stationary state and its linear stability therefore depend on β only through $u_{\rm eff}$, exactly; at the Galilean point $\beta = \alpha$ both slots vanish and every pattern translates with full symmetry.

Landau–Lifshitz form. Solving Eq. (1) for $\partial_t\mathbf{m}$ gives the equivalent condition

$$0 = -\mathbf{m}\times\mathbf{H} - \alpha\,\mathbf{m}\times(\mathbf{m}\times\mathbf{H})$$
$$+ (\mathbf{W}_1\cdot\nabla)\mathbf{m} + \mathbf{m}\times(\mathbf{W}_2\cdot\nabla)\mathbf{m},$$
$$\mathbf{W}_1 = (1+\alpha^2)\mathbf{v} - (1+\alpha\beta)\mathbf{u},$$
$$\mathbf{W}_2 = \mathbf{w}_2 + \alpha\mathbf{w}_1 = (\beta-\alpha)\mathbf{u}. \tag{4}$$

The dissipative slot of the Landau–Lifshitz form is exactly $(\beta-\alpha)\mathbf{u}$: it points along $\mathbf{u}$ whatever the Hall motion of the pattern [Fig. 2(b)]. The Gilbert pair has clean time-reversal parity and is used for the selection rules; the Landau–Lifshitz pair separates the Hall motion (all in $\mathbf{W}_1$) from the drive (all in $\mathbf{W}_2$) and is used for the stability threshold.

Coupling dictionary. The Zhang–Li terms are the antisymmetric part of a nearest-neighbour nonreciprocal coupling [Fig. 2(a)]. Along a bond $\hat{e}$ a site receives two kinds of nonreciprocal field: a field coupling κ_f that depends only on the sender, and a receiver-dependent coupling κ_r whose field also depends on the receiving spin. With the central difference $\mathbf{S}_{i+\hat{e}} - \mathbf{S}_{i-\hat{e}} = 2a\,\partial_{\hat{e}}\mathbf{S}$, term-by-term comparison with Eq. (1) gives

$$u = -2a\,\kappa_r, \qquad \beta u = -2a\,\kappa_f, \tag{5}$$

so the adiabatic term is a receiver-dependent coupling, the nonadiabatic term a field coupling, and the Galilean point is $\kappa_f = \alpha\kappa_r$. The overall sign depends only on the bond orientation convention. All numerical results below use central differences and therefore solve the lattice problem with nearest-neighbour (κ_f, κ_r) exactly. Couplings that a current cannot produce enter the same framework (Sec. VIII).

III. SELECTION RULES FROM THE RESIDUAL GROUP

The residual group of a driven pattern consists of the unitary elements g with $R_g\mathbf{w}_1 = \mathbf{w}_1$, $R_g\mathbf{w}_2 = \mathbf{w}_2$ and the antiunitary elements gT with $-R_g\mathbf{w}_1 = \mathbf{w}_1$, $R_g\mathbf{w}_2 = \mathbf{w}_2$. First-order response tensors must be covariant under it. Table I lists the magnetic point group, which decides the first-order modes, and the magnetic space group of a minimal commensurate realization, identified numerically by decorating every site with markers of $\mathbf{w}_1$ and $\mathbf{w}_2$ and analysing the result with spglib [15] (even period; orthorhombic host for one-dimensional patterns, hexagonal host for skyrmion lattices).

TABLE I. Residual magnetic groups of driven chiral patterns and the first-order modes they allow. BNS numbers refer to the minimal commensurate realization.

Pattern	Bias	Residual point group	Residual space group (BNS)	First-order mode
Néel cycloid	none	$mm21'$	$P_a ma2$ (28.92)	–
	$\mathbf{u} \parallel \mathbf{q}$	$m'm2'$	$Pm'a2'$ (28.89)	$M \parallel \hat{z} \times \mathbf{u}$
	$\mathbf{u} \perp \mathbf{q}$	$mm'2'$	$Pm'm2'$ (25.59)	none[a]
Bloch helix	none	$2221'$	P_c222_1 (17.12)	–
	$\mathbf{u} \parallel \mathbf{q}$	$2'2'2$	$P2'2'2_1$ (17.9)	$M \parallel \mathbf{u}$
	$\mathbf{u} \perp \mathbf{q}$	$2'22'$	$P2'2'2$ (16.3)	none[a]
Néel SkX	none	$6m'm'$	$P6m'm'$ (183.189)	–
	$\mathbf{w}_1 \perp$ mirror	m'	Cm' (8.34)	$\mathbf{d} \perp \mathbf{w}_1$
	generic	1	$P1$	$\mathbf{d} \perp \mathbf{w}_1$[b]
Bloch SkX	none	$62'2'$	$P62'2'$ (177.153)	–
	$\mathbf{w}_2 \parallel 2'$ axis	$2'$	$C2'$ (5.15)	$\mathbf{d} \perp \mathbf{w}_1$

[a] $(\mathbf{w}_1 \cdot \nabla)\mathbf{m} \equiv 0$, so the unbiased pattern remains an exact solution.
[b] Fixed by linear response of the unbiased group.

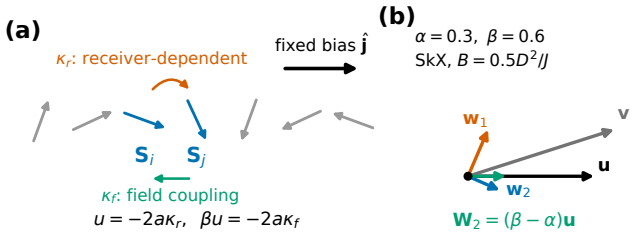


FIG. 2. A current is a fixed nonreciprocal bond bias, and in the co-moving frame it enters only through two vectors of opposite time-reversal parity. (a) On a bond $i \to j$ the adiabatic term is a receiver-dependent coupling κ_r and the β term a field coupling κ_f [Eq. (5)]. (b) Slot vectors of the driven skyrmion lattice ($B = 0.5D^2/J$, $\alpha = 0.3$, $\beta = 0.6$) from the exact Thiele projection: $\mathbf{w}_1$ (T-odd) and $\mathbf{w}_2$ (T-even) are perpendicular, and the Landau–Lifshitz drive $\mathbf{W}_2 = (\beta - \alpha)\mathbf{u}$ points along $\mathbf{u}$ whatever the Hall angle of $\mathbf{v}$.

Nonsymmorphic protection. The elements that forbid an out-of-plane cone M_z of a Néel cycloid with $\mathbf{u} \parallel \mathbf{q}$ are the glide $m_y t_{\lambda/2}$ and the screw $2'_z t_{\lambda/2}$. Both exist only because a single-harmonic pattern is invariant under a half-period translation combined with time reversal, $Tt_{\lambda/2}$; for an incommensurate pattern this is the internal translation $(1'|0, 0, 0, \frac{1}{2})$ of its magnetic superspace group, so the rule holds in general. For a pattern locked to the lattice with an odd period the half translation is lost, the group drops to Pm' (6.20), and M_z appears as a weak lock-in effect. Space-group numbers depend on the host lattice; the point group and the nonsymmorphic generators are the robust statement.

Cone response: a pattern-mediated Edelstein effect. For a harmonic helix, $\partial_q \mathbf{m} = q\,\hat{e} \times \mathbf{m}$ generates rotations about the axis $\hat{e}$, and its conjugate is the uniform cone magnetization M along $\hat{e}$. With the cone energy $e(M) = -\Delta(1 - M^2)$ the stationary condition $s u_{\rm eff} q = \partial e/\partial M$ gives, for exchange plus DMI ($q = D/J$, $2\Delta = Dq$),

$$M = \frac{s\,u_{\rm eff}}{D}. \tag{6}$$

The cone scales with $(\beta/\alpha - 1)$ and with $1/D$, and does not depend on the exchange stiffness. It points along $\hat{z} \times \mathbf{u}$ in a Néel medium and along $\mathbf{u}$ in a Bloch medium, the symmetry of the electronic Edelstein effect, and reverses with the chirality [Fig. 1(a)]. Because exchange plus DMI factorizes the cone energy as $1 - M^2$, the pitch does not change at this order.

Skyrmion lattice. Linear response of $6m'm'$ allows a dipolar distortion $\mathbf{d} = a\,\hat{z} \times \mathbf{w}_1 + b\,\mathbf{w}_2$; since $\mathbf{w}_2 \perp \mathbf{w}_1$, $\mathbf{d} \perp \mathbf{w}_1$. Ellipse axes are parallel or perpendicular to $\mathbf{w}_1$ (a 45° tilt would need $\mathrm{sym}(\mathbf{w}_1 \otimes \mathbf{w}_2)$, which m' forbids), and the lowest orientational torque is $|\mathbf{w}_1|^6 \sin 6(\theta - \phi_w)$, so a uniform current alone does not rotate the lattice [10]. Figure 3 checks the distortion rule on driven stationary states: after the uniform translation is removed, the distortion axis stays perpendicular to $\mathbf{w}_1$ within 1.5° for three values of α and three current directions, while its angle to $\mathbf{u}$ follows the Hall rotation.

IV. PATTERN WALKER BREAKDOWN: UNWINDING VERSUS THE LANDAU CRITERION

No precession. The conjugate variable of a domain wall, its tilt angle, is periodic, so its restoring force is bounded and a strong enough drive makes the wall precess [7]. The conjugate variable of a pattern, the cone M, has a quadratic energy, so its restoring force grows without bound up to $|M| = 1$ [Fig. 4(a,b)]. The static branch therefore ends in one of two ways, whichever comes first: unwinding at $|M| = 1$, $u_{\rm unw} = D/s$ for exchange plus DMI, or a Landau instability of the Goldstone mode,

$$\omega_0(k) < \lambda\,u_{\rm eff}\,k. \tag{7}$$

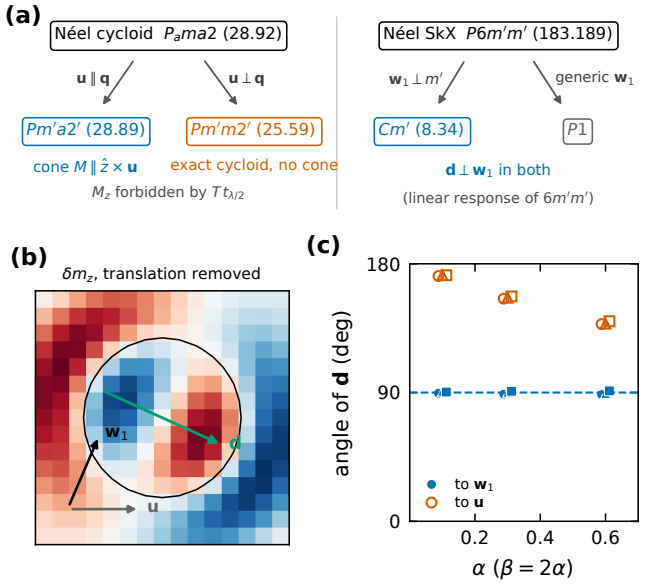


FIG. 3. The residual magnetic group fixes the first-order response, and the vector that matters is $\mathbf{w}_1$, not $\mathbf{u}$. (a) Group–subgroup steps (BNS numbers). Cycloid: $\mathbf{u} \parallel \mathbf{q}$ allows a uniform cone $M \parallel \hat{z} \times \mathbf{u}$; $\mathbf{u} \perp \mathbf{q}$ leaves the cycloid exact; M_z is forbidden by $Tt_{\lambda/2}$. Skyrmion lattice: linear response of $6m'm'$ gives $\mathbf{d} \perp \mathbf{w}_1$. (b) Change of m_z of a driven Néel skyrmion ($\alpha = 0.3$, $u = 0.02$ along x) with the translation removed; contour $m_z = 0$ of the static skyrmion; arrow the first moment $\mathbf{d}$ inside $r < 5$. (c) Angle of $\mathbf{d}$ from $\mathbf{w}_1$ (filled) and from $\mathbf{u}$ (open) for $\alpha = 0.1$, 0.3, 0.6 and current directions 0°, 30°, 90°.

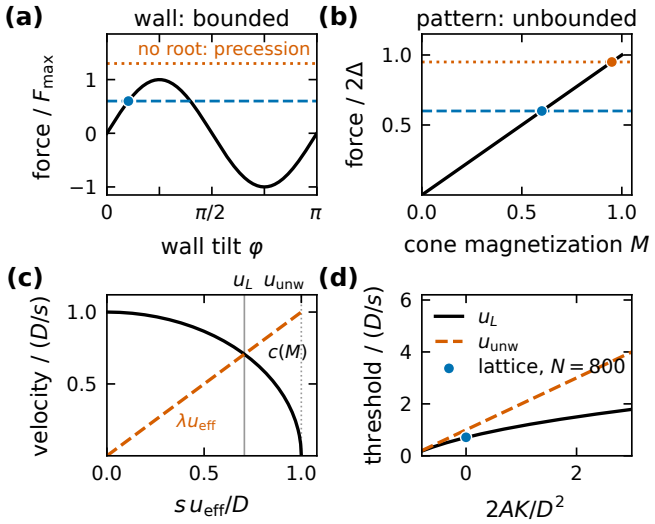


FIG. 4. A pattern has no precessional Walker breakdown; its static branch ends in a Landau instability before it can unwind. (a) The tilt of a domain wall has a bounded restoring force; a drive above its maximum (dotted) has no static root. (b) The cone M of a pattern has a linear restoring force up to $|M| = 1$. (c) Helix: the compression velocity on the driven cone meets the Doppler line at $u_L = D/\sqrt{2}s$, below $u_{\rm unw} = D/s$. (d) With easy-axis anisotropy $u_L < u_{\rm unw}$ for every K; point: exact Bloch threshold of an $N = 800$ chain.

Origin of the criterion. In the Landau–Lifshitz form a Bloch phonon Ψ_k of the co-moving background has, to first order, the growth rate

$$\mathrm{Re}\,\lambda_k = -\alpha P_k\,\omega_k + \mathbf{W}_2 \cdot \mathbf{K}_k^P, \tag{8}$$

where P_k is the ellipticity of the mode and $\mathbf{K}_k^P$ its dissipative Doppler overlap, the shift of the damping rate per unit dissipative advection. The reactive slot $\mathbf{W}_1$ only shifts the real frequency at this order. Setting $\mathrm{Re}\,\lambda_k = 0$ with $\mathbf{W}_2 = (\beta - \alpha)\mathbf{u} = \alpha u_{\rm eff}\hat{u}$ gives Eq. (7) with

$$\lambda = \kappa_P \equiv \frac{\hat{u} \cdot \mathbf{K}_k^P}{P_k k} \tag{9}$$

(Appendix A). The ellipticity cancels, so the criterion holds for overdamped modes too, and the Hall motion of the pattern does not enter because $\mathbf{W}_2 \parallel \mathbf{u}$. Damping enters only at order α^2, through the deformation of the driven background and the mixing of the phonon with internal modes (Sec. V). ω_0 is evaluated on the driven background.

Helix. For exchange plus DMI with $\mathbf{u} \parallel \mathbf{q}$ the compression phonon on the driven cone has the phase velocity $c = (D/s)\sqrt{1 - M^2}$ and $\lambda = 1$. With Eq. (6),

$$u_L = \frac{D}{\sqrt{2}\,s} < u_{\rm unw} = \frac{D}{s}\,: \tag{10}$$

the helix always breaks by the Landau instability before it unwinds [Fig. 4(c)]. With an easy-axis anisotropy K, writing $\Lambda = \sqrt{D^2 + 2AK}/s$ and $\kappa = D/\sqrt{D^2 + 2AK}$, one finds $u_L = \Lambda/\sqrt{1 + \kappa^2}$ and $u_{\rm unw} = \Lambda/\kappa$, so $u_L < u_{\rm unw}$ for every K [Fig. 4(d)].

Dispersion order decides. A linear Goldstone dispersion gives a finite threshold. A quadratic one, $\omega \simeq c_2 k^2$, gives $u_{\rm eff}^* = c_2 k/\lambda \to 0$, the reason why the Landau critical velocity of an ideal Bose gas vanishes [16]. This is the case for the gyrotropic magnetophonon of a skyrmion lattice [17] and, in the continuum, for stripe bending. In a finite sample the threshold is set by the box, $u_{\rm eff}^* \simeq 2\pi c_2/(\lambda L)$, by a pinning gap of a sliding lattice, $u_{\rm eff}^* = 2\sqrt{\omega_{\rm pin} c_2}/\lambda$, or for stripes by the orientational stiffness of the host lattice. A statically pinned lattice depins first, 20–40 times below the sliding-lattice threshold (Appendix D).

Doppler efficiency. If the phonon phase advances continuously through the texture (the affine limit), $\lambda = 1$, and Eq. (7) reduces to the criterion of Okumura *et al.* for skyrmion strings [8]. If the phase advances in steps from skyrmion to skyrmion (the rigid-particle limit), $(\mathbf{u} \cdot \nabla)$ does not act on it and $\lambda = 0$: in a rigid Thiele model the current is a constant force that drops out of the phonon equation. Real patterns lie in between, and Sec. VI shows what sets λ.

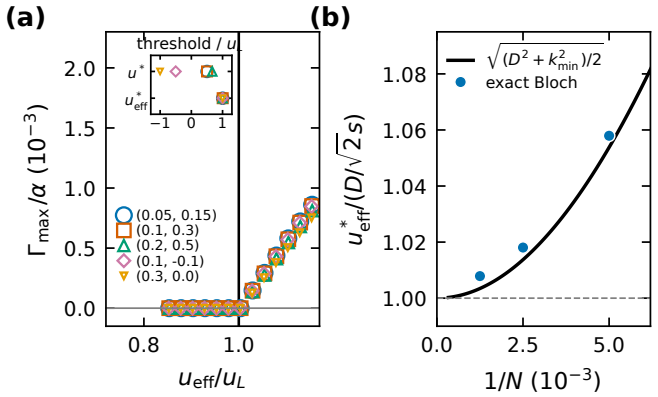


FIG. 5. The driven helix breaks at $u_L = D/\sqrt{2}s$ for every (α, β); β enters only through $u_{\rm eff}$. (a) Largest Bloch growth rate divided by α against $u_{\rm eff}$ for five (α, β) ($N = 200$; u_L includes the finite-chain correction). Inset: the same thresholds in the bare current u and in $u_{\rm eff}$. (b) Exact thresholds of finite chains (points) against $\sqrt{(D^2 + k_{\rm min}^2)/2}$ (line).

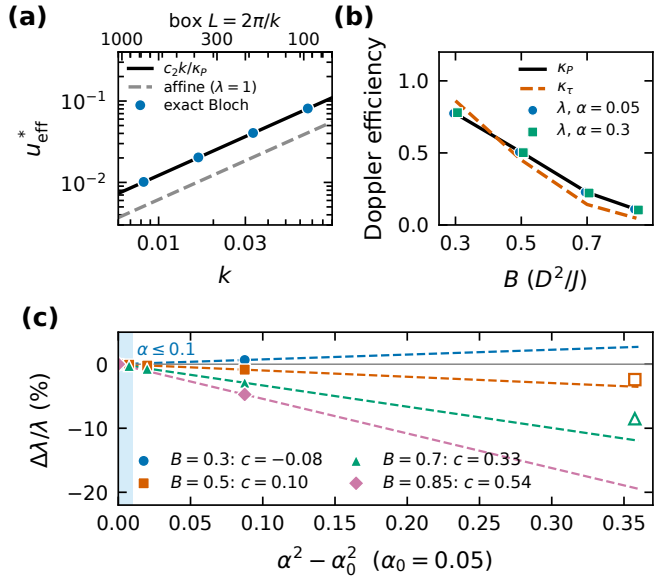


FIG. 6. The skyrmion-lattice threshold is $u_{\rm eff}^* = c_2 k/\lambda$ with $\lambda = \kappa_P$; damping enters only at order α^2. (a) Exact Bloch thresholds ($B = 0.5D^2/J$, $\alpha = 0.3$, $\beta = 0.6$) against $c_2 k/\kappa_P$ (line) and the affine estimate (dashed); top axis the box length $L = 2\pi/k$. (b) λ from exact thresholds at $\alpha = 0.05$ and 0.3 against κ_P (solid) and κ_τ (dashed). (c) $\Delta\lambda/\lambda$ against $\alpha^2 - \alpha_0^2$, fitted for $\alpha \le 0.3$ (filled); open symbols $\alpha = 0.6$; shaded $\alpha \le 0.1$.

V. TESTS: HELIX, SKYRMION LATTICE, STRIPES

All thresholds below are computed exactly from Bloch Jacobians of co-moving stationary states relaxed to a residual $\max|\dot{\mathbf{m}}| < 10^{-10}$ (one-dimensional patterns $< 10^{-12}$; for skyrmion lattices the remaining residual of $\sim 10^{-7}$ is a uniform translation from the frame velocity and changes growth rates by less than 10^{-7}).

Helix. The driven cone is exact on the lattice, $M = u_{\rm eff}\sin\theta/[2(\sqrt{J^2 + D^2} - J)] \to s u_{\rm eff}/D$, with residual 10^{-15}. For five pairs (α, β), including $\beta = 0$ and $\beta < 0$, the bare thresholds u^* spread over a factor of two and both signs, while $u_{\rm eff}^*$ agree to 0.1% [Fig. 5(a)]. Finite chains follow $\sqrt{(D^2 + k_{\rm min}^2)/2}$, the Landau criterion at the smallest allowed wave number, and converge to $D/\sqrt{2}$ [Fig. 5(b)].

Skyrmion lattice. We use a square lattice with $D = 0.4$ and a 19×33 cell holding two skyrmions. At $B = 0.5D^2/J$ the threshold is proportional to k, $u_{\rm eff}^* = 0.0101$, 0.0203, 0.0405, 0.081 at $k = 0.0083$, 0.0165, 0.033, 0.066, and agrees with $c_2 k/\kappa_P$ to 0.4–0.9% without fitting; the affine estimate $c_2 k$ is off by a factor of two [Fig. 6(a)]. Over four fields λ falls from 0.77 to 0.11 and follows the dissipative overlap κ_P, not the reactive overlap κ_τ [Fig. 6(b), Table II]. The damping dependence is quadratic, $\Delta\lambda/\lambda = -c(\alpha^2 - \alpha_0^2)$ with $c = -0.08$, 0.10, 0.33, 0.54 for $B = 0.3$, 0.5, 0.7, 0.85 [Fig. 6(c)], so for $\alpha \le 0.1$ the threshold equals $c_2 k/\kappa_P$ within 0.6%. The sign of c follows $\kappa_P - \kappa_\tau$: the leading correction is a cross term $(w_{1x}/u_{\rm eff})(\kappa_\tau - \kappa_P) = O(\alpha^2)$ between damping and the Hall residual of the reactive slot, plus deformation of the driven background and mixing with internal modes of the same order (Appendix A). At $\alpha = 0.6$ the phonon is near critical damping ($\alpha P \approx 1$), and we restrict quantitative statements to $\alpha \le 0.3$.

Real-time breakdown. The Bloch analysis predicts the dynamics directly. In a 76×66 box (16 skyrmions) at $u_{\rm eff} = 0.15$, 1.5 times the threshold of the $k_1 = 2\pi/76$ mode, the satellite $|m_z(k_1)|$ grows from noise at exactly the Bloch rate of that mode, $\Gamma = 7.26 \times 10^{-4}$, without any fitted slope, while at $u_{\rm eff} = 0.05$ the noise decays [Fig. 7(a,b)]. The growing mode is a box-length wave of skyrmion displacements along the current; once it saturates the lattice disorders without losing skyrmions. In a 76×33 box, lab-frame, Bloch and overlap thresholds agree within 4% (0.0995, 0.1016, 0.1031), while the affine estimate (0.049) is off by a factor of two and the rigid-particle picture predicts no instability at all.

Stripes. On the square lattice a stripe with $\mathbf{q} \parallel \hat{x}$ is an orientational saddle, so we use the stable diagonal orientation with periods $P = 30$, 60, 120. The bending

TABLE II. Skyrmion lattice (19×33, $k = 0.0165$, $\beta = 2\alpha$): phonon velocity ω/k, overlaps, ellipticity and the Doppler efficiency $\lambda = \omega/(k u_{\rm eff}^*)$ from exact thresholds.

B (D^2/J)	ω/k	κ_P	κ_τ	P	$\lambda_{\alpha=0.05}$	$\lambda_{\alpha=0.3}$
0.30	0.01195	0.773	0.862	1.95	0.774	0.779
0.50	0.01018	0.505	0.451	1.70	0.506	0.502
0.70	0.00588	0.226	0.142	1.58	0.228	0.222
0.85	0.00315	0.107	0.047	1.55	0.109	0.104

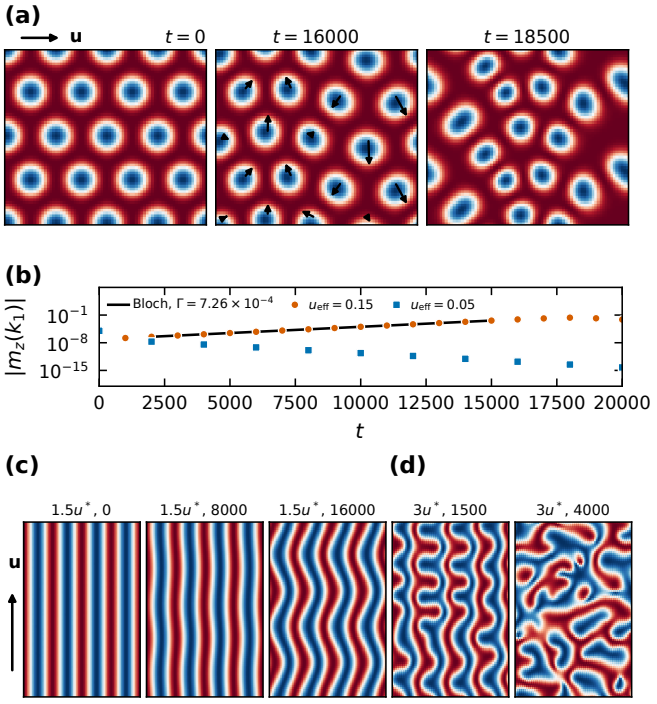


FIG. 7. Above the Doppler threshold the Goldstone mode grows at the Bloch rate and then reorganizes the pattern. (a) Skyrmion lattice at $u_{\rm eff} = 0.15$ (1.5 times threshold; 76×66, co-moving frame); arrows at $t = 16000$: skyrmion displacements since $t = 0$ ($\times 4$). (b) $|m_z(k_1)|$ against time (points) and the Bloch growth rate of the same mode (line, no fit); below threshold ($u_{\rm eff} = 0.05$) the noise decays. (c) Stripes ($P = 30$) driven along themselves at $1.5u^*$: the longest bending wave grows and saturates into a zigzag. (d) At $3u^*$: shorter waves, dislocations and a labyrinth. The late stages are treated in Ref. [13].

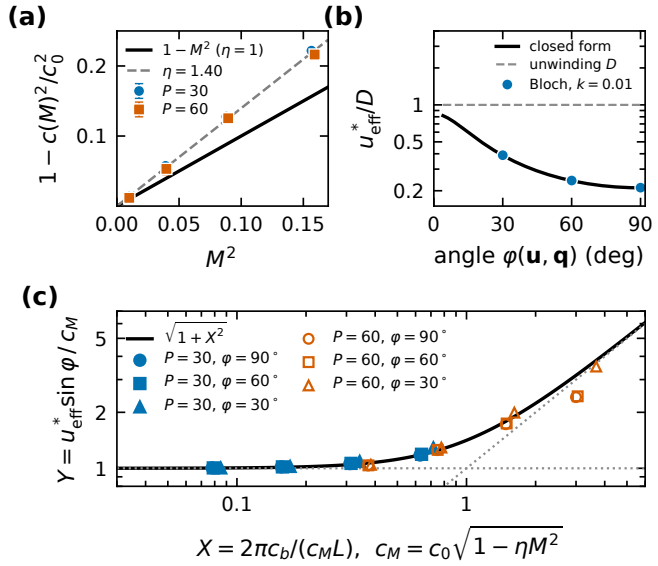


FIG. 8. Stripes have a quadratic Goldstone mode; a current along $\mathbf{q}$ only scales the bending velocity, and finite-box thresholds collapse. (a) $1 - c(M)^2/c_0^2$ against M^2 from the k-resolved ratio $\omega(k; M)/\omega(k; 0)$ (error bars: spread over k); $\eta = 1.40$ for $P = 30$ and 60. (b) Threshold against the current angle ($P = 30$): exact Bloch thresholds at $k = 0.01$ and Eq. (12). (c) Dynamical thresholds in boxes of length L, $P = 30$ (filled) and 60 (open), collapse onto $Y = \sqrt{1 + X^2}$.

branch (wave vector along the stripes) has no gap,

$$\omega^2 = c_0^2(1 - \eta M^2)k^2 + c_b^2 k^4, \tag{11}$$

where $c_0 = 0.0635$, 0.0165, 0.0048 for $P = 30$, 60, 120 vanishes as $D^{1.9}$: the linear term is a lattice-anisotropy effect, and in the continuum stripes have a quadratic Goldstone mode like the skyrmion lattice; $c_b \to 0.52$. A current component along $\mathbf{q}$ builds the cone $M = su_{\rm eff}\cos\varphi/D$ (φ is the angle between $\mathbf{u}$ and $\mathbf{q}$) and only scales the bending velocity. Measured mode by mode, the ratio $\omega(k; M)/\omega(k; 0)$ is the same for every k in the long-wave range and gives $\eta = 1.40$ for both $P = 30$ and 60 [Fig. 8(a)]. The k independence is what the conjugacy of the stripe phase and the cone requires (Appendix C); the value of η, larger than the exchange value 1, reflects the anisotropic origin of c_0. With the Doppler projection $u_{\rm eff}\sin\varphi$ on the stripe direction and $\lambda = 1$, the Landau criterion at $k \to 0$ gives

$$u_{\rm eff}^* = \frac{c_0}{\sqrt{\sin^2\varphi + \eta(c_0/D)^2\cos^2\varphi}}, \tag{12}$$

which reproduces the exact thresholds at $\varphi = 90°$, 60°, 30° [Fig. 8(b)] and tends to the unwinding point D as $\varphi \to 0$. In a box of length L along the stripes the longest bending mode sets $u_{\rm eff}^*\sin\varphi = \sqrt{c_M^2 + (2\pi c_b/L)^2}$ with $c_M = c_0\sqrt{1 - \eta M^2}$; dynamical thresholds in sheared boxes of $L = 64$–512 collapse onto this form for both periods and all three angles [Fig. 8(c)], except where the box wave number approaches the stripe wave number ($P = 60$, $L = 64$). Above threshold the longest bending wave grows first, its wave number rising as $\sqrt{u/u^* - 1}$ near threshold and $\propto u$ far above, as reported for helices [9], and saturates into a zigzag at $1.5u^*$ [Fig. 7(c,d)].

VI. WHAT SETS THE DOPPLER EFFICIENCY

Two metrics, one mode. Two exact identities of the linearized Gilbert operator J at $\alpha = 0$,

$$\frac{\partial J}{\partial \mathbf{w}_2} = -\Omega\,\frac{\partial J}{\partial \mathbf{w}_1}, \qquad \frac{\partial J}{\partial \alpha} = -\Omega J, \tag{13}$$

with Ω the symplectic form of the tangent plane (Appendix A), turn both Doppler overlaps into momenta of the same Bloch mode ψ, measured in two metrics:

$$\kappa_\tau = \frac{\langle\psi|\Omega\mathcal{P}|\psi\rangle}{k\langle\psi|\Omega|\psi\rangle}, \qquad \kappa_P = \frac{\langle\psi|\mathcal{P}|\psi\rangle}{k\langle\psi|\psi\rangle},$$
$$P = \frac{\langle\psi|\psi\rangle}{|\langle\psi|\Omega|\psi\rangle|}, \tag{14}$$

TABLE III. Static theorem and two-field theory: dynamic, static and two-field values of both overlaps for seven skyrmion lattices (cell width a, field B in D^2/J).

	κ_P			κ_τ		
a, B	dyn.	static	two-field	dyn.	static	two-field
19, 0.3	0.773	0.775	0.779	0.862	0.862	0.863
19, 0.5	0.505	0.506	0.507	0.451	0.453	0.453
19, 0.7	0.226	0.227	0.225	0.142	0.143	0.145
19, 0.85	0.107	0.107	0.111	0.047	0.047	0.050
13, 0.5	0.228	0.229	0.234	0.104	0.105	0.104
15, 0.5	0.327	0.327	0.332	0.217	0.219	0.218
23, 0.6	0.339	0.340	0.337	0.282	0.284	0.287

where $\mathcal{P}$ is the advection (momentum) generator. κ_τ is the momentum in the symplectic (Berry) metric and sets the reactive Doppler shift; κ_P is the momentum in the dissipative metric and sets the threshold.

Static theorem. Write the Bloch mode as $\Psi_k = e^{ikx}(\phi_G + k\phi_1)$, with ϕ_G the Goldstone mode. λ is fixed by the corrector ϕ_1. The identity $[H_0, x] = -i\partial_k H_k$ makes $\phi_1 = -ix\phi_G$ an exact solution except at cell boundaries, so $\lambda = 0$ when ϕ_G vanishes between pattern units (the rigid limit) and $\lambda = 1$ when the texture is uniform. For a type-B Goldstone mode ($\omega \propto k^2$) the dynamical term is $O(k^2)$, so the $O(k)$ corrector solves a static problem: both overlaps equal those of the statically relaxed strain mode and are independent of the dynamics and of α. Table III confirms this for seven lattices.

One-dimensional theorem. If the conjugate field of the in-plane angle θ is slaved locally by a strong easy-plane anisotropy, a ground-state transformation maps the problem onto homogenization with conductivity $\sigma = \psi_0^2$, $\psi_0 = \theta_0'$, and

$$\lambda = \frac{1}{\langle\psi_0^2\rangle\langle\psi_0^{-2}\rangle} \tag{15}$$

(with a nonuniform conjugate stiffness $M(x)$, $\langle M^{-1}\rangle/[\langle\psi_0^{-2}\rangle\langle\psi_0^2/M\rangle]$). A uniform helix gives $\lambda = 1$; a dilute soliton lattice gives $\lambda \propto e^{-L/w}$ with $w = \sqrt{J/B}$. For a chiral soliton lattice [18] Eq. (15) agrees with the exact overlap within 2% at nine fields, and fails as it should when the conjugate field is not slaved [Fig. 9(a)].

Counterflow. Figure 10 shows what the static corrector does. Under a long-wave strain the affine mode squeezes and stretches the skyrmions with the lattice; the relaxed mode with the same skyrmion displacements keeps them round, and the difference between the two is a pattern of spin canting at the skyrmion cores. This canting carries a Berry-phase momentum that cancels part of the Doppler momentum of the affine mode: at a wavelength of two cells the momentum per k drops from 0.95 to 0.72 at $B = 0.3$ and from 0.93 to 0.07 at $B = 0.85$. Hence, with $\Pi_{\rm cf}$ the counterflow momentum (per unit affine momentum) carried by the internal relaxation,

$$\lambda = 1 - \Pi_{\rm cf}, \tag{16}$$

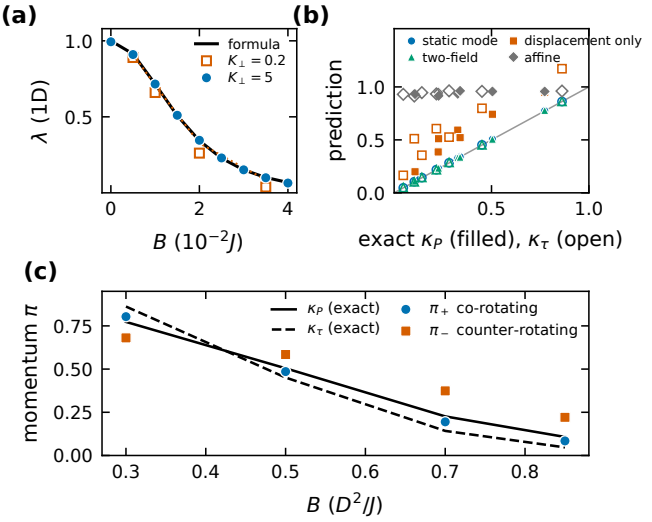


FIG. 9. λ is a static property: one minus the counterflow momentum of the internal relaxation. (a) Chiral soliton lattice: exact overlap against Eq. (15) with a slaved conjugate field ($K_\perp = 5$); with $K_\perp = 0.2$ the formula (dotted) overestimates λ. (b) Seven skyrmion lattices: predictions against exact κ_P (filled) and κ_τ (open). (c) Momenta of the co-rotating (π_+) and counter-rotating (π_-) components with the exact overlaps.

the analogue of Feynman backflow or of a $k \cdot p$ effective mass. A two-field Ritz theory with the displacement envelope and its spin-conjugate canting envelope, without fitted parameters, reproduces both overlaps of all seven lattices to ± 0.01, while displacement-only theories and the affine mode do not [Table III, Fig. 9(b)]. The canting is only 4–16% of the corrector norm but carries a large momentum, because λ is a cross term between displacement and canting; the counterflow is a magnetic, not an elastic, effect. Projecting the corrector on local channels shows that the helicity twist is always rigid, because DMI locks the helicity, and that λ follows the affinity of the elliptic (shear) channel. Soft ellipse modes reverse the counterflow and give $\lambda > 1$ (1.69 at $a = 23$, $B = 0.4$).

Why $\kappa_P > \kappa_\tau$ in dilute lattices. Split ψ into its co-rotating and counter-rotating tangent-plane components (eigenvectors of Ω) with momenta per norm $\pi_\pm$. The dissipative metric adds the two components and the symplectic metric subtracts them (Appendix G). The counterflow cancels mainly the co-rotating, gyrotropic momentum, $\pi_+ = 0.80 \to 0.08$ for $B = 0.3 \to 0.85$, while the counter-rotating admixture keeps more, $\pi_- = 0.68 \to 0.22$ [Fig. 9(c)], so $\kappa_P > \kappa_\tau$ exactly when $\pi_- > \pi_+$.

VII. BELOW THRESHOLD: A MOVING GIBBS ENSEMBLE

Inserting the Doppler term into the dynamics of the gyrotropic displacement field gives an effective energy $E - \mathbf{v}\cdot\mathcal{P}$ with $\mathbf{v} = \lambda u_{\rm eff}\hat{u}$. Its pseudo-momentum term is

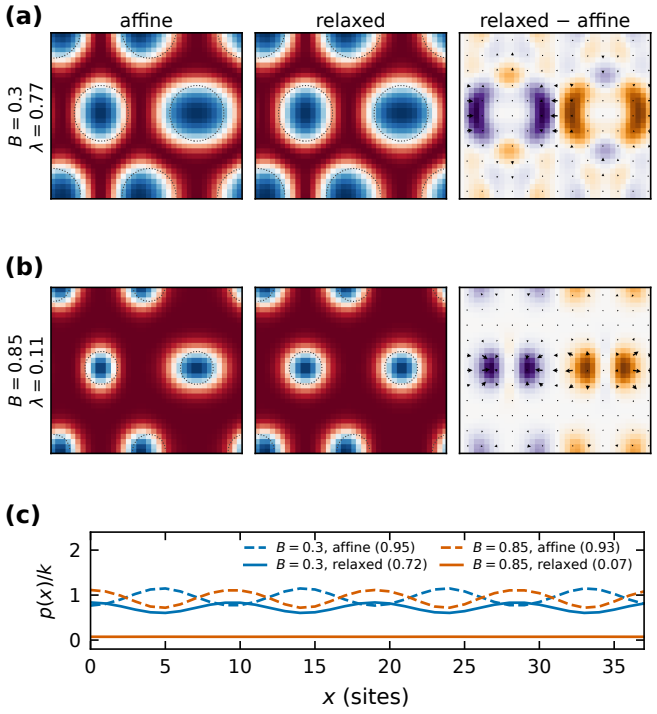


FIG. 10. λ in real space. (a, b) A longitudinal strain wave (wavelength two cells, displacement amplitude 2 sites, exaggerated) at $B = 0.3D^2/J$ ($\lambda = 0.77$) and $0.85D^2/J$ ($\lambda = 0.11$). Left: affine mode; middle: statically relaxed mode with the same displacement; right: their difference, the internal canting (colour δm_z, arrows in-plane). Dotted: $m_z = 0$ of the undeformed lattice. (c) Doppler-momentum density of the phonon averaged across the strip, divided by k; averages in brackets.

a Lifshitz invariant of the displacement field with coefficient $(G/2)\lambda(\beta-\alpha)u/\alpha$ (Appendix B), a "DMI of the lattice displacement". A phonon carries pseudo-momentum $\pm k$, so the two branches at the same k carry effective temperatures

$$T_{\rm eff} = \frac{T}{1 \mp x}, \qquad x = \frac{\lambda u_{\rm eff} k}{\omega_k}, \tag{17}$$

and the Landau threshold $x = 1$ is where this ensemble stops being normalizable. Because the displacement field is real, the branches (k, ω) and $(-k, \omega)$ are conjugate and the damping rates obey the sum rule $\Gamma_+ + \Gamma_- = 2\Gamma_0$ to first order; numerically $\Gamma_+ + \Gamma_- = 0.834 + 2.551 = 3.385$ against $2\Gamma_0 = 3.388$ (same units) at k_1 and $u_{\rm eff} = 0.05$, and the deviations grow as u^2.

Stochastic LLG simulations at $k_BT = 0.01$ test Eq. (17). At $k_1 = 2\pi/76$ the two branches follow $1/(1\mp x)$ up to $x = 0.7$ [Fig. 11(a)]. In a 152×66 box at $u_{\rm eff} = 0.03$, modes along the current heat as $1/(1-x^2)$ while transverse and oblique modes stay at the bath temperature [Fig. 11(b)]; since $x \propto k_\parallel/k^2$, heating concentrates in the longest wavelengths along the current.

Melting. Heating ladders of 256–264 skyrmions give $T_m(0) = 0.201 \pm 0.002$, close to the KTHNY estimate

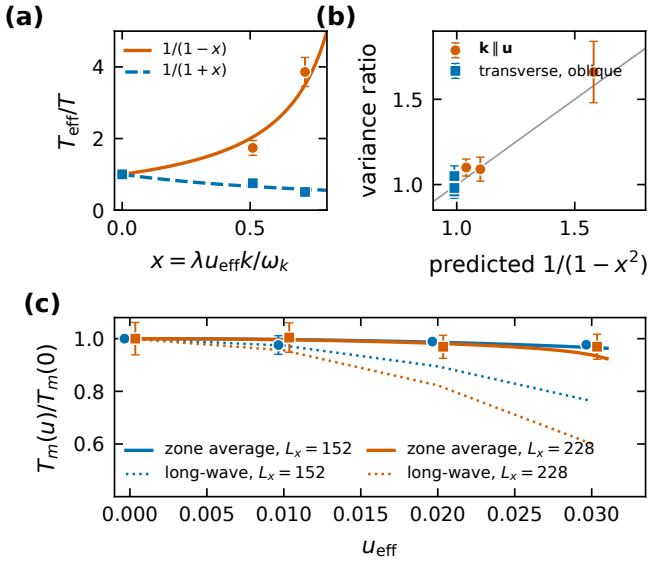


FIG. 11. Below threshold the current puts the lattice phonons in a Gibbs ensemble moving at $\lambda u_{\rm eff}$, and the melting point moves only by the zone average of that heating. (a) Effective temperatures of the two branches at $k_1 = 2\pi/76$ (76×33, $k_BT = 0.01$) against $1/(1\mp x)$. (b) Variance ratio of displacement modes (152×66, $u_{\rm eff} = 0.03$) against $1/(1-x^2)$. (c) Melting temperature against $1/\langle(1-x^2)^{-1}\rangle_{\rm BZ}$ (solid) and the long-wave criterion (dotted) for two box shapes.

from the elastic constants [19, 20]. A current below threshold barely moves it, $T_m(u)/T_m(0) = 0.97$–1.00 within errors of 0.01–0.05, although the longest mode of the larger box is already at $x = 0.89$ [Fig. 11(c)]. Melting is controlled by bond-scale strain fluctuations, to which every mode contributes equally by equipartition, so the relevant measure is the Brillouin-zone average

$$\frac{\Delta\langle\varepsilon^2\rangle}{\langle\varepsilon^2\rangle} = \Big\langle \frac{1}{1-x_k^2} \Big\rangle_{\rm BZ} - 1 \approx \Big(\frac{\lambda u_{\rm eff}}{c_2}\Big)^2 \frac{\pi \ln(k_{\max}/k_{\min})}{A_{\rm BZ}}, \tag{18}$$

which is 3.5% and 6.6% for the two boxes at $u_{\rm eff} = 0.03$ and agrees with the measured shifts; a criterion based on the longest modes predicts 24–40% and is ruled out. Because a lattice below threshold has $u_{\rm eff} \le c_2 k_{\min}/\lambda$, Eq. (18) is bounded by $\pi A_{\rm cell} \ln(Lk_{\max}/2\pi)/L^2$, which vanishes for large L: below the Doppler threshold a current cannot lower the melting temperature of a large sample appreciably. Far above threshold the current melts the lattice into a flowing skyrmion liquid [13].

VIII. BEYOND CURRENT: NONRECIPROCAL DMI AND DAMPING-LIKE SPIN–ORBIT TORQUE

Exchange-parity rule. A first-order nonreciprocal DMI has $\mathbf{n}_{ji} = \mathbf{n}_{ij}$, so it is even in the bond direction. The neighbour sum $\sum_{\hat e} \mathbf{n}\times(\mathbf{S}_{i+\hat e}+\mathbf{S}_{i-\hat e}) \approx \sum_{\hat e} \mathbf{n}\times(2\mathbf{S}_i+\partial_{\hat e}^2\mathbf{S})$ then has no first-gradient term: an exchange-odd cou-

pling (δJ) advects, which is the Zhang–Li case, while an exchange-even one (δD) acts locally plus through a second-order nonreciprocal stiffness. Its time-reversal-even channel is a damping-like torque $-\mathbf{S}\times(\mathbf{S}\times\mathbf{n})$, i.e., damping-like spin–orbit torque (SOT) [21]; its odd channel is a field-like torque. In a Néel medium the map $\mathbf{n}\leftrightarrow\hat{z}\times\mathbf{p}$ gives a field-like $\mathbf{n}$ the residual group of $\mathbf{w}_1$ and a damping-like $\mathbf{n}$ that of $\mathbf{w}_2$.

Damping-like SOT pumps with zero threshold. The field-like torque neither moves the lattice nor pumps (damping rates change by less than 0.05%). The damping-like torque ($c=0.004$, $\mathbf{n}=\hat{y}$) makes the lattice flow and pumps modes with $\mathbf{k}$ close to $\hat{z}\times\mathbf{n}$, with $x\cdot k$ nearly constant, so $x\propto 1/k$ and the threshold vanishes in an infinite system [Fig. 12(a)]. The mechanism is not advection: pure drift cancels the Doppler term because its dissipative slot vanishes. An exact first-order decomposition shows that the pumping is a strain-dependent force; applied to the static relaxed mode the SOT generator reproduces the exact pumping coefficient at every k, and applied to the affine mode it gives exactly zero [Fig. 12(b)].

Duality. Zhang–Li acts fully on the affine part, and the internal relaxation reduces it ($\lambda<1$). Damping-like SOT gives zero on the affine part, and the internal relaxation alone produces the pumping. Both are overlaps of a nonreciprocal generator on the same static relaxed mode, so λ and the SOT pumping coefficient are static material quantities of the same kind. At equal lattice velocity SOT pumps about 1.7 times more strongly than the current.

Second-order nonreciprocal stiffness. The full exchange-even coupling contains, beyond its local Zeeman-like part, the stiffness $T_2=\delta D\,P_\perp(\mathbf{n}\times\nabla^2\mathbf{S})$. It is odd in $\mathbf{S}$, hence reactive, and non-variational, so unlike a translation-invariant energy it exerts a net force on a pattern, $\mathbf{F}\parallel\mathbf{n}$ for a Néel lattice (Appendix F); the Thiele response $(G-\alpha\mathcal{D})\mathbf{v}=\mathbf{F}$ reproduces the measured drift to three digits. On a modulated background ∇^2 acts on a phonon envelope as $2i\mathbf{k}\cdot\nabla$, so T_2 contains a k-linear Doppler term although the coupling is even in the bond direction: for $\mathbf{k}\perp\mathbf{n}$ a frequency shift that damping converts into a splitting, for $\mathbf{k}\parallel\mathbf{n}$ an intrinsic splitting carried by the static relaxed mode. Both splittings are $\propto k$ while the bare damping is $\propto k^2$, so the threshold $\delta D^*\propto k$ vanishes [Fig. 12(c)]: a second route to zero-threshold pumping that needs no current.

IX. WHERE THE MODEL MEETS REAL SAMPLES

The Landau criterion (7) rests on two assumptions: the adiabatic torque has the structure of advection, and damping acts as friction against the lattice. Both follow from the form of the Zhang–Li torque and of Gilbert damping, not from microscopic details, so the threshold is set by the ratio of $u_{\rm eff}$ to the DMI velocity D/s and by

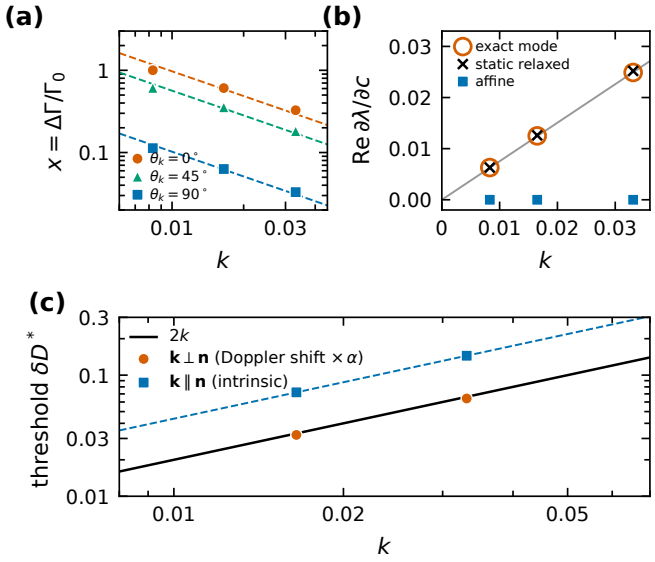


FIG. 12. Exchange-even couplings pump with zero threshold through the static relaxed mode, dual to the Zhang–Li Doppler term. (a) Damping-like SOT ($c=0.004$, $\mathbf{n}=\hat{y}$, $\alpha=0.3$): pumping ratio $x=\Delta\Gamma/\Gamma_0$ for three directions of $\mathbf{k}$; dashed $1/k$. (b) First-order SOT generator on the exact, static relaxed and affine modes ($\alpha=0$). (c) Nonreciprocal DMI stiffness: threshold δD^* for $\mathbf{k}\perp\mathbf{n}$ and $\mathbf{k}\parallel\mathbf{n}$; both $\propto k$.

two properties of the pattern, the order of its Goldstone dispersion and λ. Real magnets have $\alpha\lesssim 0.1$, where $\lambda=\kappa_P$ within 0.6%, so λ can be computed from the static pattern alone. The velocity scale is $D/s=\gamma D/M_s$; with $u_{\rm eff}=(\beta/\alpha-1)u$ and $u=g\mu_B Pj/(2eM_s)$, the helix threshold $u_L=D/\sqrt{2}s$ corresponds to

$$j_L=\frac{\sqrt{2}\,e\gamma D}{g\mu_B P}\,\frac{\alpha}{\beta-\alpha}, \tag{19}$$

independent of M_s and of the exchange stiffness. The predictions, written as dependencies, are:

- The cone magnetization $M/M_s=su_{\rm eff}/D$ grows with $\beta/\alpha-1$ and $1/D$, along $\hat{z}\times\mathbf{j}$ in Néel and along $\mathbf{j}$ in Bloch media, and reverses with the chirality, consistent with the sign rule observed in MnP [11].
- The helix breaks at $j_L\propto D\alpha/(\beta-\alpha)$, Eq. (19), a factor $1/\sqrt{2}$ below the point where the cone would close [9].
- Skyrmion lattices flow stably in small samples because their threshold rises as $1/L$; in weakly pinned sliding lattices it is $2\sqrt{\omega_{\rm pin}c_2}/\lambda$.
- The skyrmion-lattice threshold rises with field as c_2/λ, because skyrmions become particle-like and λ falls.
- The threshold is insensitive to damping and to the skyrmion Hall angle at fixed $u_{\rm eff}$; changes are of order α^2.

- Stripes are destabilized by any current component along them, with a threshold that vanishes as $1/L$, the longest bending wave going first; a component along $\mathbf{q}$ only softens the bending velocity by $\sqrt{1-\eta M^2}$.
- Distortions and ellipse axes of a driven skyrmion lattice lock to the Hall-rotated $\mathbf{w}_1$, not to $\mathbf{j}$.
- Below threshold the longest phonons along the current are hot, $T_{\text{eff}} = T/(1-x)$, visible in their spectra, while the melting temperature moves only by the zone average, a few percent.

X. CONCLUSION

A nonreciprocal spin coupling with a fixed direction, whether produced by a current or designed, acts on a chiral pattern through one structure: two vectors of opposite time-reversal parity in the co-moving stationary equation, whose dissipative combination is exactly $(\beta-\alpha)\mathbf{u}$. This structure fixes which distortions a driven pattern may develop, rules out a precessional Walker breakdown and replaces it with a Doppler instability of the pattern's own Goldstone mode. Whether a pattern survives a finite current is decided by the order of its Goldstone dispersion; how strongly the current couples to that mode is decided by a static property, the Doppler efficiency λ, which measures how continuously the pattern phase advances and is set by the counterflow of strain-induced spin canting. Exchange-even couplings reach the same modes through the dual route. What a pattern does beyond its threshold is the subject of Ref. [13].

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The code and data that reproduce every figure are available at https://github.com/cywon-commits/NonReciprocalInteraction_SelfOrganizedPattern (branch `b1/request6`, folders `figs_b1` and `results/b1_figs`).

Appendix A: Operator identities and the Landau criterion

Linearize the Gilbert equation about a co-moving stationary state in the local tangent frame with symplectic form Ω ($\Omega^2 = -1$), and write $J(\mathbf{w}_1, \mathbf{w}_2, \alpha)$ for the Bloch Jacobian at wave vector k. Because $\mathbf{m}\times$ acts on the tangent plane as $-\Omega$ and the Gilbert form gives $\partial_t \mathbf{m} = (\mathbf{T} + \alpha \mathbf{m}\times\mathbf{T})/(1+\alpha^2)$, the identities (13) hold exactly at $\alpha = 0$; we verified them numerically to 10^{-10}. For a Hamiltonian Bloch mode $J\psi = i\omega\psi$, first-order perturbation theory with left vector $\Omega^T\psi$ gives

$$\delta\lambda = -i\, w_{1x} k\kappa_\tau + w_{2x} P k \kappa_P - \alpha P \omega. \tag{A1}$$

The reactive slot shifts only the frequency, and damping then acts on the shifted frequency. Writing $\mathbf{w}_2 = \mathbf{W}_2 - \alpha\mathbf{w}_1$ with $\mathbf{W}_2 = \alpha u_{\text{eff}}\hat{x}$ gives

$$\begin{aligned} \operatorname{Re}\lambda &= \alpha P\big[\lambda u_{\text{eff}} k - \omega\big], \\ \lambda &= \kappa_P + \frac{w_{1x}}{u_{\text{eff}}}(\kappa_\tau - \kappa_P) + O(\alpha^2)_{\text{bg}}. \end{aligned} \tag{A2}$$

The exact Thiele solution gives $w_{1x} = \alpha\mathcal{D}^2(\beta-\alpha)u/(G^2+\alpha^2\mathcal{D}^2)$, so the cross term is of order α^2; "bg" marks terms of the same order from the deformation of the driven background and from second-order mixing with internal modes. The leading term $\lambda = \kappa_P$ is independent of α, of β beyond u_{eff}, and of the Hall angle.

Appendix B: Moving Gibbs ensemble

Project the stochastic LLG on the pumped mode a_k. With the growth rate of Appendix A and a bath noise fixed by Gilbert damping (fluctuation–dissipation at temperature T for the undriven rate $\Gamma_0 = \alpha P\omega$),

$$\dot{a}_k = \big[i\omega_k - \Gamma_0(1-x)\big]a_k + \xi_k, \qquad \langle \xi_k \xi_k^* \rangle = 2\Gamma_0 \frac{T}{\varepsilon_k}, \tag{B1}$$

so $T_{\text{eff}} = T/(1-x)$, and the conjugate branch has $1+x$. The stationary distribution is $\exp[-(E - \mathbf{v}\cdot\mathcal{P})/T]$ with $\mathbf{v} = \lambda u_{\text{eff}}\hat{x}$ and $\mathcal{P}$ the pseudo-momentum of the gyrotropic displacement field $\mathbf{u}(\mathbf{r})$. For a type-B mode $\varepsilon_k/\omega_k = G|u_k|^2$, so

$$E - \mathbf{v}\cdot\mathcal{P} = E - \frac{G}{2}\lambda\frac{\beta-\alpha}{\alpha} u \int d^2r\, (u_x\partial_x u_y - u_y\partial_x u_x), \tag{B2}$$

a Lifshitz invariant of the displacement field. The derivation needs the noise projection to be unchanged at $O(u)$, which holds for $\alpha \le 0.3$.

Appendix C: Stripe bending threshold

With Eq. (11) and the Doppler projection $u_{\text{eff}}\sin\varphi$, $\min_k \omega/k$ is reached at $k \to 0$, at $c_0\sqrt{1-\eta M^2}$ with $M = s u_{\text{eff}}\cos\varphi/D$; solving gives Eq. (12). The stripe phase θ and the cone M are conjugate, and for exchange plus DMI the cone energy $-\Delta(1-M^2)$ has the M-independent curvature 2Δ. A phase stiffness that is quadratic in the rotating amplitude $r = \sqrt{1-M^2}$ then gives $\omega^2 = (r^2 K_\theta k^2)(2\Delta)/s^2$: every such phase

velocity scales by the same factor, independent of k, as the helix compression velocity $(D/s)\sqrt{1-M^2}$ does. The lattice bending stiffness c_0^2 comes from the orientational anisotropy of the host lattice, whose dependence on r need not be quadratic; the measured factor is k-independent, as the conjugacy requires, with $\eta = 1.40$. In the continuum $c_0 \to 0$, and a box of length L along the stripes gives $u^*_{\rm eff} \sin\varphi = 2\pi c_b/L$.

Appendix D: Pinning

A weak pinning potential adds $K_{\rm pin}$ to both elastic eigenvalues of the displacement field, so $\omega(k) \approx \omega_{\rm pin} + c_2 k^2$ and the Landau minimum at $k^* = \sqrt{\omega_{\rm pin}/c_2}$ gives $u^*_{\rm eff} = 2\sqrt{\omega_{\rm pin} c_2}/\lambda$. This assumes a sliding lattice. A pinned lattice must first depin: for a commensurate potential with spring constant $K = \omega_{\rm pin}|G|$ the maximum pinning force is $Ka/2\pi$, and the current pushes a static lattice with $|F| = u\sqrt{G^2 + \beta^2\mathcal{D}^2}$, so

$$u_{\rm dp} = \frac{\omega_{\rm pin}\, a\, |G|}{2\pi\sqrt{G^2 + \beta^2\mathcal{D}^2}}, \qquad \frac{u^*_{\rm eff}}{u_{\rm dp}} \propto \omega_{\rm pin}^{-1/2}. \tag{D1}$$

For a lattice with a pinning field $\delta B = \epsilon B$ that follows m_z, fully converged stationary states with their static deformation confirm both pieces: for $\epsilon = 0.02$ (0.08) the lattice depins between $u = 0.002$ and 0.003 (0.008 and 0.016), against $u_{\rm dp} = 0.0025$ (0.0096), stays linearly stable until then, and the sliding-lattice threshold is 0.106 (0.213). A pinned lattice therefore depins 20–40 times below its Doppler threshold.

Appendix E: Direction of SOT pumping

For damping-like SOT with $\mathbf{n} = \hat{y}$, the branch splitting at $\mathbf{k} \parallel \hat{y}$ is 0.106 of that at $\mathbf{k} \parallel \hat{x}$, independent of the torque strength. The pumping axis is tilted by 6° from $\hat{z} \times \mathbf{n}$ while the lattice flows at 67°; the axis is set by the strain-dependent force tensor, whose antisymmetric part is about 10% of its symmetric part, not by the drift.

Appendix F: Force of the nonreciprocal stiffness

With $T_2 = \delta D\, P_\perp(\mathbf{n} \times \nabla^2\mathbf{S})$ and $\mathbf{S}\cdot\nabla^2\mathbf{S} = -|\nabla\mathbf{S}|^2$, the Thiele force $F_\mu = \int \partial_\mu\mathbf{S}\cdot(\mathbf{S}\times T_2)$ becomes, after one integration by parts, with $\sigma_{\mu\nu} = \partial_\mu\mathbf{S}\cdot\partial_\nu\mathbf{S}$,

$$F_\mu = \delta D \int d^2r \Big[(\mathbf{n}\cdot\partial_\nu\mathbf{S})\sigma_{\mu\nu} - \tfrac{3}{2}(\mathbf{n}\cdot\partial_\mu\mathbf{S})\,\mathrm{tr}\,\sigma\Big] \tag{F1}$$

In a Néel skyrmion $\mathbf{n}\cdot\mathbf{S} = \sin\theta\,(\mathbf{n}\cdot\hat{r})$, so $\mathbf{F} \parallel \mathbf{n}$; in a Bloch skyrmion $\mathbf{F} \parallel \hat{z}\times\mathbf{n}$. For the test lattice ($B = 0.5D^2/J$) the exact lattice force is $6.81\,\delta D\,\mathbf{n}$ and the closed form gives $6.57\,\delta D\,\mathbf{n}$; the anisotropy of σ carries 40% of it. $(G - \alpha\mathcal{D})\mathbf{v} = \mathbf{F}$ gives $\mathbf{v} = (0.238, -0.105)\delta D$ for $\mathbf{n} = \hat{y}$, equal to the integrated drift.

Appendix G: Circular decomposition of the overlaps

Let $\psi = \psi_+ + \psi_-$ be the decomposition into the $\pm i$ eigenspaces of Ω, with $n_\pm = \langle\psi_\pm|\psi_\pm\rangle$ and $\pi_\pm = \langle\psi_\pm|\mathcal{P}|\psi_\pm\rangle/(kn_\pm)$. The advection generator commutes with Ω, so $\mathcal{P}$ does not mix the two components. With $r = n_-/n_+ = (P-1)/(P+1)$,

$$\kappa_P = \frac{\pi_+ + r\pi_-}{1+r}, \qquad \kappa_\tau = \frac{\pi_+ - r\pi_-}{1-r},$$
$$\kappa_P - \kappa_\tau = \frac{2r(\pi_- - \pi_+)}{1-r^2}. \tag{G1}$$

On the lattice the cross term $\langle\psi_+|\mathcal{P}|\psi_-\rangle$ is 0.2–1.5% of the diagonal, and the split reproduces both overlaps within 1% (Table IV).

TABLE IV. Circular decomposition of the Bloch mode (19×33, $k = 0.0165$).

B	P	π_+	π_-	π_-/π_+	κ_P	κ_τ
0.30	1.95	0.804	0.681	0.85	0.774	0.862
0.50	1.70	0.486	0.584	1.20	0.506	0.451
0.70	1.58	0.195	0.374	1.92	0.228	0.142
0.85	1.55	0.084	0.221	2.62	0.108	0.047

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